

ORIGINAL ARTICLE

Ten-Second Heart Rate Variability, Its Changes Over Time, and the Development of Hypertension

Jeonggyu Kang, Yoosoo Chang, Yejin Kim, Hocheol Shin, Seungho Ryu^{ID}

BACKGROUND: The role of ultrashort-term heart rate variability (HRV) and its temporal changes in incident hypertension are unknown. We aimed to investigate the association between 10-second HRV, its changes, and incident hypertension in adults aged <40 years and older.

METHODS: This cohort study included 232 587 Koreans (mean age 37.6 years) without hypertension. Hypertension was defined according to the 2017 American College of Cardiology and American Heart Association hypertension guidelines. HRV, including the root mean square of successive R-R interval differences and the SD of normal-to-normal R-R intervals, was estimated using standard 12-lead, 10-second electrocardiography.

RESULTS: During a median follow-up of 3.8 years, 40 268 hypertension cases were identified (incidence rates: 36.1 and 67.9 per 1000 person-years for young and older participants, respectively). An inverse association was observed between HRV and hypertension risk, in a dose-dependent manner. The multivariable-adjusted hazard ratios (95% CIs) for hypertension comparing the first to the fifth quintiles of root mean square of successive R-R interval difference and SD of normal-to-normal R-R interval were 1.58 (1.52–1.63) and 1.35 (1.30–1.39), respectively. These associations were stronger in young adults than in older adults. In a subsample of 150 301 participants, compared with stable HRV, an increase in HRV over time was also inversely associated with incident hypertension.

CONCLUSIONS: A higher HRV and its increase over time on a 10-second electrocardiography were associated with a lower risk of hypertension. Our findings indicate that autonomic function, estimated using 10-second standard electrocardiography, plays a role in predicting hypertension, with a stronger effect in young adults. (*Hypertension*. 2022;79:1308–1318. DOI: 10.1161/HYPERTENSIONAHA.121.18589.) • [Supplemental Material](#)

Key Words: blood pressure ■ cohort studies ■ electrocardiography ■ heart rate ■ incidence ■ young adult

Hypertension is a major risk factor for stroke, coronary heart disease, heart failure, kidney disease, and premature mortality worldwide.¹ Although hypertension is more prevalent in older individuals, its prevalence in the younger population aged <40 years is reportedly up to 18% and is expected to increase in parallel with obesity rates.² Moreover, a younger age of hypertension onset is associated with higher risks of cardiovascular disease (CVD) and mortality.³ Therefore, it is important to implement strategies to prevent hypertension and identify its risk factors, even in young age groups with fewer

comorbidities. Although increased peripheral vascular resistance is the hallmark of hypertension, an altered autonomic nervous system contributes to increased blood pressure (BP). In particular, the early stages of hypertension in young individuals with borderline hypertension were characterized by hyperkinetic features derived from sympathetic overactivity.⁴

The heart rate variability (HRV) test is a noninvasive, feasible, and the most widely used test in clinical practice for the assessment of autonomic nervous system activity. Although short-term HRV must be measured

Correspondence to: Seungho Ryu, Department of Occupational and Environmental Medicine, Kangbuk Samsung Hospital, School of Medicine, Sungkyunkwan University, Samsung Main Bldg B2, 250, Taepyeong-ro 2ga, Jung-gu, Seoul 04514, South Korea, Email sh703.yoo@gmail.com or Yoosoo Chang, Department of Occupational and Environmental Medicine, Kangbuk Samsung Hospital, School of Medicine, Sungkyunkwan University, Samsung Main Bldg B2, 250, Taepyeong-ro 2ga, Jung-gu, Seoul 04514, South Korea, Email yoosoo.chang@gmail.com

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NOVELTY AND RELEVANCE

What Is New?

Ultrashort heart rate variability (HRV) from 10-second standard electrocardiography data is related to hypertension risk among young (<40 years) and older individuals, particularly young adults.

Temporal changes in 10-second HRV were related to hypertension risk.

What Is Relevant?

Ten-second HRV and its temporal changes were independent risk factors for incident hypertension and may help identify high-risk individuals.

Clinical/Pathophysiological Implications?

Ten-second HRV can stratify and monitor young and older adults at high risk for incident hypertension.

These results may inform studies on the relationship between ultrashort HRV measured using wearable devices and hypertension or other outcomes.

Nonstandard Abbreviations and Acronyms

CVD	cardiovascular disease
HRV	heart rate variability
hs-CRP	high-sensitivity C-reactive protein
RMSSD	root mean square of successive R-R interval differences
SBP	systolic blood pressure
SDNN	SD of normal-to-normal R-R intervals

for at least 5 minutes, recent studies using 10-second, 12-lead electrocardiography have shown associations between ultrashort HRV and cardiac and noncardiac events.^{5,6} Furthermore, changes in HRV are associated with cardiovascular risk factors and can be induced by aging, environmental factors, or intentional behavioral modifications such as weight loss and exercise.^{7–10} To our knowledge, no studies have evaluated the role of HRV changes in the development of hypertension, especially in a young population.

Therefore, our study aimed to determine whether baseline 10-second HRV derived from 12-lead electrocardiography and its temporal changes are associated with hypertension in a large sample of healthy young and middle-aged adults.

METHODS

The data supporting the findings of this study and analytical methods are available from the corresponding author upon reasonable request. The data are not shared publicly because we do not have permission from the institutional review board to distribute them.

Study Population

This study was performed as part of the Kangbuk Samsung Health Study, a cohort study of Korean adults aged ≥18 years who underwent annual or biennial health examinations in

Seoul or Suwon, South Korea (further details in [Supplemental Material](#)).^{11,12} This study was restricted to participants who underwent a health examination from 2011 (the year in which raw data of 10-second electrocardiography recordings were available) to 2017 and received at least one follow-up examination by December 2019 (n=308 428).

First, we excluded 75 841 participants who met one or more of the following criteria at baseline ([Figure S1](#)): missing information on systolic BP (SBP) or diastolic BP, body mass index, the SD of normal-to-normal R-R intervals (SDNN), or the root mean square of successive R-R interval differences (RMSSD) derived from 10-second electrocardiography; history of hypertension, use of antihypertensive agents, or BP ≥130/80 mm Hg; history of CVD, including any heart disease or stroke; current use of antiarrhythmic drugs; current use of antidepressants or psychoactive drugs; and premature atrial or ventricular contractions, atrial fibrillation or flutter, second- or third-degree atrioventricular block, and other supraventricular or ventricular arrhythmias on electrocardiography. A total of 232 587 participants were included in the analysis of the relationship between baseline HRV and hypertension incidence.

For the analysis of the association between serial changes in HRV and incident hypertension, participants with data on baseline and subsequent visits but missing data on at least one follow-up visit were also excluded. Participants with missing data on HRV or BP and those who were diagnosed with hypertension at the second visit were also excluded, leaving 150 301 participants.

This study was approved by the Institutional Review Board of Kangbuk Samsung Hospital (Institutional Review Board no. 2020-07-046), which waived the requirement for informed consent, as we only used deidentified data routinely collected as part of the health screening examinations.

Measurements

Information on medical and family history, medication use, lifestyle factors, and education level was obtained through a self-administered questionnaire; BP and anthropometric and serum biochemical parameters were measured by trained staff at baseline and follow-up visits (further details in [Supplemental Material](#)).^{13–15}

Trained nurses measured BP using an automated oscillometric device (53000, Welch Allyn, New York) with the

participants in a sitting position with the arm supported at the level of the heart. Three consecutive BP readings were obtained after the participants had rested quietly in a sitting position for 5 minutes. The average of the second- and third-BP readings was used in the analysis. Hypertension was defined as SBP ≥ 130 mm Hg, diastolic BP ≥ 80 mm Hg, self-reported history of hypertension, or use of antihypertensive medication.¹⁶ The threshold of 140/90 mm Hg and above was used for diagnosis of stage 2 hypertension.

HRV Measurements

A standard 12-lead electrocardiography was performed as a routine part of the health screening in all participants on the same day that BP was measured. Electrocardiography was obtained with a 500-Hz sampling frequency and amplification of 1 mV/cm using an electrocardiography recorder (CARDIMAX FX-7542; Fukuda Denshi Co, Ltd, Tokyo, Japan; further details in [Supplemental Material](#)). These digitalized electrocardiography data were also analyzed using an automated electrocardiography analysis system (PI-19E; Fukuda Denshi Co, Ltd, Tokyo, Japan), which automatically detected the QRS complex and calculated R-R intervals. Among the simultaneously recorded 12-lead signals, lead II was used to calculate the HRV due to its significantly larger R-peak and common usage in HRV measurements.¹⁷ The 2 time-domain HRV parameters, including SDNN and RMSSD, were calculated on the basis of the 10-second electrocardiography recording. The heart rate was derived from the average R-R intervals. In a subsample of the study population ($n=48\,111$) whose both 10-second electrocardiography and 3-minute HRV were measured on the same day, the intraclass correlation coefficient value was 0.532 (95% CI, 0.265–0.551) for SDNN and 0.708 (0.706–0.713) for RMSSD between 10-second and 3-minute HRV.

Statistical Analyses

Descriptive statistics were used to summarize the characteristics of the participants based on incident hypertension. SDNN and RMSSD were then categorized into quintiles based on their own distributions (further details in [Supplemental Material](#)), as there is no clear established cutoff value for either measurement to define autonomic dysfunction. The highest quintile was chosen as the reference group because low HRV is associated with cardiovascular events.^{18–20} The changes in HRV were calculated for each participant as the difference in HRV (either RMSSD or SDNN) between the baseline and subsequent visit (visit 1 and visit 2) values. These changes were categorized into quintiles of distribution, and the category containing zero (no-change group) was chosen as the reference.

The primary end point was hypertension development. Person-years were calculated as the sum of the follow-up duration from baseline until the development of hypertension or until the final examination conducted before December 31, 2019, whichever occurred first. The incidence rate was calculated as the number of incident cases divided by the person-years of follow-up. Cox proportional hazards models were used to estimate adjusted hazard ratios and 95% CIs for incident hypertension, comparing each HRV quintile 1–4 (separately) to quintile 5 (as the reference).

The statistical models were initially adjusted for age and sex, and then further adjusted for year of screening examination,

center, alcohol consumption, smoking, physical activity, total energy intake, sleep duration, medication for hyperlipidemia, education level, depressive symptoms, history of diabetes, and body mass index. To evaluate the effects of changes in HRV and other covariates during the follow-up period, we performed analyses that introduced HRV and other factors as time-varying covariates in the models.

Stratified analyses were performed according to age (<40 versus ≥ 40 years). The effects of interactions between age group and HRV categories on incident hypertension were tested using likelihood ratio tests that compared models with and without multiplicative interaction terms.

All statistical analyses were performed using STATA version 16.0 (Stata Corp LP, College Station, TX). Statistical significance was set at $P<0.05$.

RESULTS

The baseline characteristics of the study participants according to the incidence of hypertension are shown in Table 1. The mean (SD) age was 37.6 (7.5) years, and the median SDNN and RMSSD (interquartile range) were 28.0 (19.6–40.1) and 34.5 (23.4–50.4), respectively. Incident hypertension was positively associated with age, male sex, current smoking, alcohol intake, physical activity, and a higher prevalence of diabetes and antihyperlipidemic medication use. Furthermore, incident hypertension was positively associated with unfavorable lipid profiles and higher body mass index, BP, glucose levels, and hs-CRP (high-sensitivity C-reactive protein) levels.

During 877 728 person-years of follow-up, 40 268 participants developed hypertension (incidence rate: 45.9 per 1000 person-years) over a median follow-up period of 3.8 years (interquartile range, 2.0–5.7 years). The HRV quintiles, including the RMSSD and SDNN quintiles, were significantly and inversely associated with an increased risk of hypertension in a dose-response manner (P for trend <0.001 ; Tables 2 and 3). After adjustment for age, sex, center, year of screening examination, alcohol consumption, smoking status, physical activity, total energy intake, sleep duration, medication for hyperlipidemia, education level, depression, history of diabetes, and body mass index, the multivariable-adjusted hazard ratios (95% CIs) for incident hypertension in RMSSD quintiles 4, 3, 2, and 1 compared with the highest quintile were 1.15 (1.11–1.19), 1.24 (1.20–1.29), 1.35 (1.30–1.40), and 1.58 (1.52–1.63), respectively. These associations were consistently observed in time-dependent analyses. Similar trends were also consistently observed in analyses using SDNN, another index of HRV, instead of RMSSD. After further adjustment for either resting heart rate (Model 2, Tables 1 and 2) or glucose, SBP lipid, and hs-CRP levels, the associations between RMSSD, SDNN, and incident hypertension were attenuated but remained significant (Table S1).

In stratified analyses by age, the absolute incidence rates of hypertension were lower in young individuals

Table 1. Baseline Characteristics Based on Incident Hypertension (n=232 587)

Characteristics	Overall	No hypertension	Incident hypertension	P value
N	232 587	192 319	40 268	
Age,* y	37.6 (7.5)	37.1 (7.4)	39.8 (7.5)	<0.001
Men, %	47.0	40.9	76.3	<0.001
Obesity,† %	21.4	17.5	39.8	<0.001
Current smoker, %	19.3	16.4	33.1	<0.001
Alcohol intake,‡ %	15.4	33.0	18.5	<0.001
HEPA, %	15.2	14.8	17.3	<0.001
High education level,§ %	84.4	84.4	84.2	0.296
Depressive symptoms, %	11.8	12.4	9.4	<0.001
Diabetes, %	2.0	1.6	4.0	<0.001
Medication for hyperlipidemia, %	1.1	1.0	1.7	<0.001
Heart rate,* beats/min	64.6 (8.5)	64.3 (8.4)	65.7 (8.9)	<0.001
SDNN,¶ ms	28.0 (19.6–40.1)	28.8 (20.3–41.0)	24.4 (16.9–35.1)	<0.001
RMSSD,¶ ms	34.5 (23.4–50.4)	35.8 (24.5–52.0)	28.5 (19.2–41.9)	<0.001
Sleep duration,* h	6.6 (1.1)	6.6 (1.2)	6.4 (1.1)	<0.001
Body mass index, kg/m ²	22.7 (3.1)	22.3 (3.0)	24.4 (3.1)	<0.001
Systolic BP,* mmHg	104.7 (9.8)	103.1 (9.4)	112.6 (7.8)	<0.001
Diastolic BP,* mmHg	66.3 (6.8)	65.1 (6.5)	72.2 (5.0)	<0.001
Glucose,* mg/dL	93.1 (11.7)	92.3 (10.9)	96.5 (14.5)	<0.001
Total cholesterol,* mg/dL	190.7 (33.0)	188.8 (32.5)	199.8 (33.9)	<0.001
LDL-C,* mg/dL	117.2 (31.1)	115.1 (30.6)	127.4 (31.5)	<0.001
HDL-C,* mg/dL	60.3 (15.3)	61.5 (15.3)	54.5 (14.0)	<0.001
Triglyceride,¶ mg/dL	82.0 (60.0–119.0)	78.0 (58.0–111.0)	109.0 (77.0–158.0)	<0.001
hs-CRP,¶ mg/L	0.4 (0.2–0.8)	0.4 (0.2–0.7)	0.5 (0.3–1.0)	<0.001
Total calorie intake,¶ # kcal/d	1512 (1150–1909)	1490 (1127–1886)	1615 (1270–2010)	<0.001

Data are expressed as percentage. BP indicates blood pressure; HDL-C, high-density lipoprotein cholesterol; HEPA, health-enhancing physically active; hs-CRP, high-sensitivity C-reactive protein; LDL-C, low-density lipoprotein cholesterol; RMSSD, root mean square of successive differences in R-R intervals; and SDNN, SD of normal-to-normal R-R intervals.

*Data are expressed as mean (SD).

†≥25 kg/m².

‡≥20 g/d.

§≥College graduate.

||≥16 Center for Epidemiological Studies Depression Scale score.

¶Data are expressed as median (interquartile range).

#Among 161 258 participants with plausible estimated energy intake levels (within 3 SDs from the log-transformed mean energy intake).

(aged <40 years) than in older individuals (incidence rate: 36.1 versus 67.9 per 1000 person-years for young and older adults, respectively); however, the relative association between HRV quintiles and hypertension was stronger in younger individuals than in older individuals (P for interaction=0.005 and 0.004 for RMSSD and SDNN, respectively). In the spline regression models, the incidence of hypertension decreased across the range of HRV; while this pattern was observed in young and older individuals, the association was stronger in young individuals (Figure).

Tables 4 and 5 show the relationships between HRV changes over time and hypertension. Compared with stable RMSSD (RMSSD change: quintile 3) over time, the highest increase in RMSSD (RMSSD change: quintile 5) was significantly associated with a lower risk of incident hypertension. The corresponding hazard ratios

(95% CIs) comparing RMSSD change from quintiles 5 to 3 (stable group) were 0.86 (0.81–0.91) in the young group and 0.91 (0.86–0.97) in the older group (Table 4, Model 1; P for interaction=0.307 by age group). These associations were stronger in the time-dependent analyses than in the original analyses. The inverse association between SDNN change and incident hypertension was similarly observed in both young and older individuals (P for trend <0.001), without significant interaction by age (P for interaction=0.626). After further adjustments for glucose level, lipid levels, SBP, and hs-CRP level, the association between HRV changes and hypertension remained consistent (Table S2).

The change in HRV between the baseline and subsequent visits was inversely associated with baseline HRV and showed the highest baseline HRV values in the first quintile of HRV change (Table S3). Compared

Table 2. Development of Hypertension by RMSSD Quintiles (n=232 587)

RMSSD quintiles	PY	Incident cases	Incidence density (/10 ³ PY)	Age- and sex-adjusted HR (95% CI)	Multivariable-adjusted HR* (95% CI)		HR (95% CI)† in a model with time-dependent variables
					Model 1	Model 2	
Total							
Q1 (<21.1)	161 162	12 324	76.5	1.82 (1.76–1.88)	1.58 (1.52–1.63)	1.16 (1.12–1.21)	1.51 (1.46–1.57)
Q2 (21.1–29.7)	174 420	9101	52.2	1.48 (1.43–1.53)	1.35 (1.30–1.40)	1.12 (1.08–1.16)	1.30 (1.26–1.35)
Q3 (29.8–39.7)	179 319	7550	42.1	1.31 (1.26–1.35)	1.24 (1.20–1.29)	1.09 (1.05–1.13)	1.21 (1.17–1.26)
Q4 (39.8–55.2)	180 857	6380	35.3	1.18 (1.14–1.22)	1.15 (1.11–1.19)	1.06 (1.02–1.10)	1.14 (1.10–1.18)
Q5 (≥55.3)	181 970	4913	27.0	1.00 (reference)	1.00 (reference)	1.00 (reference)	1.00 (reference)
<i>P</i> value for trend				<0.001	<0.001	<0.001	<0.001
Age <40 y							
Q1 (<21.1)	82 475	5060	61.4	2.04 (1.95–2.13)	1.74 (1.67–1.82)	1.32 (1.26–1.38)	1.70 (1.62–1.78)
Q2 (21.1–29.7)	109 123	4680	42.9	1.58 (1.52–1.65)	1.44 (1.38–1.51)	1.22 (1.16–1.27)	1.40 (1.33–1.47)
Q3 (29.8–39.7)	125 081	4389	35.1	1.36 (1.30–1.42)	1.30 (1.24–1.36)	1.15 (1.10–1.20)	1.26 (1.20–1.32)
Q4 (39.8–55.2)	138 876	4250	30.6	1.23 (1.18–1.29)	1.20 (1.15–1.25)	1.12 (1.07–1.17)	1.18 (1.12–1.24)
Q5 (≥55.3)	152 584	3585	23.5	1.00 (reference)	1.00 (reference)	1.00 (reference)	1.00 (reference)
<i>P</i> value for trend				<0.001	<0.001	<0.001	<0.001
Age ≥40 y							
Q1 (<21.1)	78 687	7264	92.3	1.72 (1.62–1.82)	1.52 (1.43–1.61)	1.20 (1.13–1.27)	1.58 (1.50–1.67)
Q2 (21.1–29.7)	65 297	4421	67.7	1.37 (1.29–1.45)	1.27 (1.19–1.35)	1.10 (1.04–1.17)	1.32 (1.25–1.40)
Q3 (29.8–39.7)	54 238	3161	58.3	1.22 (1.15–1.30)	1.17 (1.10–1.25)	1.06 (1.00–1.13)	1.22 (1.15–1.30)
Q4 (39.8–55.2)	41 980	2130	50.7	1.09 (1.02–1.17)	1.07 (1.00–1.15)	1.02 (0.95–1.09)	1.12 (1.05–1.19)
Q5 (≥55.3)	29 387	1328	45.2	1.00 (reference)	1.00 (reference)	1.00 (reference)	1.00 (reference)
<i>P</i> value for trend				<0.001	<0.001	<0.001	<0.001

P=0.005 for the overall interaction between age and RMSSD quintiles for hypertension (multivariable-adjusted Model 1). HR indicates hazard ratio; PY, person-years; and RMSSD, root mean square of successive differences in R-R intervals.

*Estimated from Cox proportional hazards models. Model 1 was adjusted for age, sex, center, year of screening examination, alcohol consumption, smoking, physical activity, body mass index, total energy intake, sleep duration, education level, depression, medication for hyperlipidemia, and history of diabetes. Model 2 was adjusted for variables included in Model 1 and heart rate.

†Estimated from Cox proportional hazard models with quintiles of RMSSD, smoking, alcohol consumption, physical activity, total energy intake, body mass index, sleep duration, depression, medication for hyperlipidemia, and history of diabetes as time-dependent categorical variables, and baseline age, sex, center, year of screening examination, and education level as time-fixed variables.

with the stable HRV category (quintile 3), the decrease category was significantly associated with a lower incidence of hypertension in the age- and sex-adjusted models, although it was associated with a higher risk of hypertension after further adjustments for baseline HRV and other confounders (Table S4 and Tables 4 and 5). The corresponding hazard ratios (95% CIs) comparing the RMSSD change from quintiles 1 to 3 (stable group) were 1.19 (1.11–1.26) in the young group and 1.21 (1.13–1.30) in the older group (Table 4, Model 1). In sensitivity analyses using stage 2 hypertension as an outcome, an association between 10-second HRV and incident hypertension was consistently observed (Tables S5 and S6).

DISCUSSION

In this longitudinal study of a large cohort of young and middle-aged participants, SDNN and RMSSD derived from 10-second conventional electrocardiography, the main time-domain parameters of HRV, were inversely associated with the risk of hypertension in a

dose-response manner. The risk of incident hypertension based on HRV categories was greater in individuals aged <40 years than in those aged ≥40 years. Furthermore, the risk of hypertension decreased in participants with the greatest increase in HRV over time, an association that remained significant after adjustment for baseline HRV.

HRV has been widely used as an index for cardiovascular risk stratification for decades. Several studies have shown that abnormal autonomic function, estimated by HRV, increases the risk of cardiovascular events and mortality in the general population and in the high-risk group.^{21,22} Reduced HRV is independently associated with CVD risk factors, such as diabetes, dyslipidemia, smoking, and obesity.^{9,23–25} Dysregulation of autonomic function has also been implicated in the development of hypertension, and evidence suggests that reduced HRV is linked to an increased risk of hypertension.^{26–29} In the Framingham Heart Study, a cross-sectional analysis revealed that several measurements of both time- and frequency-domain indices of HRV from 2 hours of ambulatory heart rate recordings were significantly lower

Table 3. Development of Hypertension by SDNN Quintiles (n=232 587)

SDNN quintiles	PY	Incident cases	Incidence density (/10 ³ PY)	Age- and sex-adjusted HR (95% CI)	Multivariable-adjusted HR* (95% CI)		HR (95% CI)† in a model with time-dependent variables
					Model 1	Model 2	
Total							
Q1 (<17.9)	162 928	11 448	70.3	1.55 (1.50–1.60)	1.35 (1.30–1.39)	1.07 (1.04–1.11)	1.26 (1.22–1.30)
Q2 (17.9–24.4)	175 150	8781	50.1	1.30 (1.26–1.35)	1.20 (1.16–1.24)	1.05 (1.01–1.08)	1.15 (1.11–1.19)
Q3 (24.5–32.0)	179 028	7623	42.6	1.20 (1.16–1.25)	1.14 (1.11–1.18)	1.04 (1.01–1.08)	1.11 (1.07–1.15)
Q4 (32.1–43.7)	179 990	6646	36.9	1.11 (1.07–1.15)	1.08 (1.04–1.12)	1.02 (0.99–1.06)	1.06 (1.02–1.10)
Q5 (≥43.8)	180 632	5770	31.9	1.00 (reference)	1.00 (reference)	1.00 (reference)	1.00 (reference)
<i>P</i> value for trend				<0.001	<0.001	<0.001	<0.001
Age <40 y							
Q1 (<17.9)	83 327	4530	54.4	1.71 (1.64–1.78)	1.47 (1.41–1.54)	1.17 (1.12–1.23)	1.70 (1.62–1.78)
Q2 (17.9–24.4)	109 930	4520	41.1	1.40 (1.34–1.46)	1.28 (1.23–1.34)	1.12 (1.08–1.17)	1.40 (1.33–1.47)
Q3 (24.5–32.0)	125 848	4465	35.5	1.25 (1.20–1.30)	1.19 (1.14–1.24)	1.09 (1.04–1.13)	1.26 (1.20–1.32)
Q4 (32.1–43.7)	138 647	4297	31.0	1.12 (1.07–1.17)	1.09 (1.05–1.14)	1.03 (0.99–1.08)	1.18 (1.12–1.24)
Q5 (≥43.8)	150 387	4152	27.6	1.00 (reference)	1.00 (reference)	1.00 (reference)	1.00 (reference)
<i>P</i> value for trend				<0.001	<0.001	<0.001	<0.001
Age ≥40 y							
Q1 (<17.9)	79 601	6918	86.9	1.54 (1.46–1.63)	1.35 (1.27–1.42)	1.13 (1.07–1.20)	1.58 (1.50–1.67)
Q2 (17.9–24.4)	65 220	4261	65.3	1.24 (1.17–1.32)	1.14 (1.08–1.21)	1.04 (0.98–1.10)	1.32 (1.25–1.40)
Q3 (24.5–32.0)	53 181	3158	59.4	1.16 (1.10–1.24)	1.11 (1.05–1.18)	1.04 (0.98–1.11)	1.22 (1.15–1.30)
Q4 (32.1–43.7)	41 343	2349	56.8	1.10 (1.03–1.17)	1.07 (1.00–1.14)	1.03 (0.97–1.10)	1.12 (1.05–1.19)
Q5 (≥43.8)	30 245	1618	53.5	1.00 (reference)	1.00 (reference)	1.00 (reference)	1.00 (reference)
<i>P</i> value for trend				<0.001	<0.001	<0.001	<0.001

P=0.004 for the overall interaction between age and SDNN quintiles for hypertension (multivariable-adjusted Model 1). HR indicates hazard ratio; PY, person-years; and SDNN, SD of normal-to-normal R-R intervals.

*Estimated from Cox proportional hazards models. Model 1 was adjusted for age, sex, center, year of screening examination, alcohol consumption, smoking, physical activity, body mass index, total energy intake, sleep duration, education level, depression, medication for hyperlipidemia, and history of diabetes. Model 2 was adjusted for variables included in Model 1 and heart rate.

†Estimated from Cox proportional hazard models with quintiles of SDNN, smoking, alcohol consumption, physical activity, total energy intake, body mass index, sleep duration, depression, medication for hyperlipidemia, and history of diabetes as time-dependent categorical variables, and baseline age, sex, center, year of screening examination, and education level as time-fixed variables.

in patients with hypertension.²⁶ However, in a previous prospective analysis, only the low-frequency power was associated with the development of hypertension in men; none of the HRV parameters were associated with the development in women.²⁶ In the Atherosclerosis Risk in Communities (ARIC) study of 7099 participants without hypertension at baseline,²⁸ a lower SDNN and RMSSD on 2-minute and 6-minute HRV recordings separated by 9 years were inversely associated with the risk of hypertension over 9 years of follow-up. Although the involvement of the autonomic nervous system in the pathogenesis of hypertension has consistently been reported, there is a discrepancy between studies across different measures of HRV, sex, and temporal association in HRV and hypertension in a population that consists primarily of young adults, which has yet to be clearly established. Our study demonstrated that in a large, asymptomatic population of young adults, decreased HRV was associated with a significantly increased risk of hypertension. In addition, we found that an increase in HRV over time was a protective marker of incident hypertension. To date, no study has explored the effects of HRV changes on hypertension.

Interventions, such as stress reduction, weight loss, exercise, smoking cessation, and some medications, are effective in increasing HRV, and these behavioral changes reduce the risk of CVD.^{9,30,31} Our results showed that lifestyle and pharmacological interventions focused on restoring sympathovagal balance and increasing HRV may be an effective prevention strategy against early-onset hypertension. Future studies are needed to determine whether the modification of health behaviors is associated with improved parameters of short-term HRV and whether its change affects prognosis.

Traditionally, 5 minutes of continuous electrocardiography recording is considered standard for short-term HRV analysis.³² However, short-term electrocardiography recordings of 30 to 60 seconds can also provide a robust estimate of HRV and maintain independent prognostic value in CVD outcomes.³³ In addition, previous studies documented the utility of ultrashort-term electrocardiography, such as 10-second electrocardiography, which is routinely available in clinical practice.³² Ultrashort-term electrocardiography has also been linked to various clinically important CVD outcomes. HRV derived